

**DESIGN AND IMPLEMENTATION OF A VISION-
GUIDED AUTONOMOUS SURVEILLANCE ROVER FOR
INTELLIGENT SECURITY AND E-MONITORING**

A PROJECT REPORT

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BONAFIDE CERTIFICATE

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ABSTRACT

The project identifies the limitations of traditional surveillance systems such as fixed infrastructure and human dependency. It clearly establishes the need for an advanced and automated monitoring solution. The main objective of the project is to develop a vision-guided autonomous surveillance rover. This ensures improved mobility and intelligent monitoring without continuous human control. The system uses a microcontroller as the central control unit for processing and coordination. It manages sensor input, decision-making, and motor control efficiently. A camera module is integrated to capture real-time visual data from the surroundings. This enables continuous monitoring and remote surveillance capability. An ultrasonic sensor is used for obstacle detection and distance measurement. It helps the rover navigate safely by avoiding collisions in real time. The motor control mechanism allows the rover to move autonomously in different directions. This allows users to monitor the area remotely without physical presence.

The methodology involves proper hardware integration and implementation of control logic. Real-time processing of sensor data enables quick decision-making during operation. This improves the efficiency and responsiveness of the surveillance system. The experimental results confirm successful navigation and obstacle detection. The system also maintains stable image transmission under normal conditions. The project highlights minor limitations such as dependency on network availability. However, the overall system performance remains reliable and consistent. The system is scalable and can be upgraded with advanced features like AI-based detection. It provides a strong foundation for future intelligent surveillance technologies.

TABLE OF CONTENTS

CONTENT	PAGE #
ABSTRACT	iv
LIST OF TABLES	vii
LIST OF FIGURES	viii
LIST OF ABBREVIATIONS	ix
CHAPTER 1 INTRODUCTION	1
1.1 Background and Motivation	1
1.2 Problem Statement	2
1.3 Objectives and Outcomes	3
1.4 Scope and Applications	5
CHAPTER 2 RELATED WORK (OR) STATE OF THE ART AND NEED FOR THE PROJECT	7
2.1 Existing Research, Technologies or Products	7
2.2 Gaps and Challenges in Existing Approaches	9
2.3 Need for the Proposed Project	11
CHAPTER 3 REQUIREMENTS AND CONSTRAINTS	12
3.1 Requirements	12
3.1.1 <i>Functional Requirements</i>	13
3.1.2 <i>Performance Requirements</i>	14
3.2 Engineering Standards and Code	15
3.3 Design constraints	17
3.3.1 <i>Economic constraints</i>	18
3.3.2 <i>Environmental and Sustainability Constraints</i>	19
3.3.3 <i>Manufacturability Constraints</i>	21
3.3.4 <i>Ethical and Health & Safety Constraints</i>	23
3.3.5 <i>Social Constraints</i>	25
3.3.6 <i>Political/Regulatory Constraints</i>	27
CHAPTER 4 DESIGN AND IMPLEMENTATION	30
4.1 Concept Development and Alternatives	30
4.1.1 <i>Concept A – Manual controlles surveillance Rover</i>	31
4.1.2 <i>Concept B – Sensor – based autonomous rover</i>	31
4.1.3 <i>Concept C – Vision – Guided autonomous surveillance rover</i>	31
4.1.4 <i>Evaluation Criteria</i>	32
4.2 Final Design and Justification	32
4.3 System Architecture	34
4.3.1 <i>Subsystem 1- Vsion and processing unit</i>	35
4.3.2 <i>Subsystem 2 -Motion and control unit</i>	35
4.4 Implementation	36
4.5 Project Management	37
4.5.1 <i>Work Breakdown Structure</i>	39
4.5.2 <i>Gantt Chatt / Timeline</i>	40
4.5.3 <i>Budget and Bill of Materials</i>	42
4.5.4 <i>Risk Assessment</i>	42
CHAPTER 5 TESTING AND EVALUATION	44
5.1 Test Plans and Procedure	44

5.2	Results	45
5.3	Evaluation Against Requirements	46
CHAPTER 6 SUMMARY AND CONCLUSION		47
6.1	Technical Achievements and Trade-offs	47
6.2	Ethical and Professional Responsibilities	47
6.3	Societal, Environmental and Safety Impacts	47
6.4	Sustainability and Global Relevance	48
6.4.1	<i>SDG Indicators</i>	48
6.4.2	<i>Relevancy of the SDG(s) to the work</i>	48
6.5	Limitations	49
6.6	Future Work Recommendations	49
6.7	Conclusion	49
APPENDICES		50
IMAGE VIEW OF PROJECT		50
PROGRAM OF PROJECT		51
REFERENCES		53

LIST OF TABLES

Table 4.1 Comparison of Alternative Solutions	33
Table 4.2 Trade-off Analysis of Design Alternatives	33
Table 4.3 A Sample Work Breakdown Table	39
Table 4.4 Team Responsibility Matrix	40
Table 4.5 Gantt Chart	40
Table 4.6 Budget of the Project	42
Table 4.7 Risk Assessment Matrix of Project	43

LIST OF FIGURES

Figure 4.1 Sample Figure for System Architecture	34
Figure 4.2 Sample Process Chart for Software	34
Figure 4.3 A Sample Work Breakdown Structure	39
Figure 4.4 Gantt Chart	41
Figure 4.5 Risk Matrix	42

LIST OF ABBREVIATIONS

ACRONYM	DEFINITION
ESP32	Microcontroller with built-in Wi-Fi capability
ESP32-CAM	Camera module based on ESP32 for video streaming
IoT	Internet of Things
DC	Direct Current
PWM	Pulse Width Modulation
Wi-Fi	Wireless communication technology
GPIO	General Purpose Input Output
L298N	Motor driver module for controlling DC motors
Vcc	Supply Voltage
GND	Ground Connection
IDE	Integrated Development Environment
PCB	Printed Circuit Board
IP	Internet Protocol
HTTP	Hypertext Transfer Protocol
FPS	Frames Per Second

CHAPTER 1 INTRODUCTION

1.1 Background and Motivation

The increasing demand for effective security systems has highlighted the limitations of traditional surveillance methods, which rely on fixed cameras and manual monitoring, leading to restricted coverage and delayed responses [1]. With advancements in embedded systems and Internet of Things (IoT) technologies, intelligent and automated surveillance solutions have emerged. Devices such as ESP32 and ESP32-CAM enable real-time monitoring and wireless data transmission, improving system efficiency [2]. This project is motivated by the need to develop a vision-based autonomous rover that provides mobile, real-time surveillance with minimal human intervention.

The rapid growth of smart technologies and connected systems has significantly influenced the development of modern surveillance solutions. Conventional systems lack flexibility and are often limited to fixed locations, making them ineffective in dynamic and large-scale environments. In contrast, mobile robotic platforms equipped with embedded systems provide enhanced adaptability and coverage. By integrating sensors and communication modules, these systems can continuously monitor surroundings, detect changes, and respond accordingly. This shift towards intelligent surveillance systems ensures improved efficiency, reduced human effort, and better real-time decision-making capabilities.

Furthermore, the integration of wireless communication technologies has enabled seamless data transmission and remote accessibility in surveillance applications. IoT-enabled devices allow users to monitor live data from anywhere, ensuring continuous supervision without physical presence. The use of microcontrollers such as ESP32 plays a crucial role in managing data processing, sensor interaction, and communication between system components. Additionally, camera modules like ESP32-CAM enhance the system by providing visual monitoring capabilities, making the surveillance process more reliable and effective in identifying potential threats or unusual activities.

In addition to monitoring capabilities, autonomous navigation is an essential feature in modern surveillance systems. The use of sensors such as ultrasonic modules enables the system to detect obstacles and navigate safely in different environments.

1.2 Problem Statement

Existing surveillance systems are largely static and depend on fixed cameras and manual monitoring, leading to limited coverage and reduced efficiency [1]. They lack autonomous capabilities such as real-time navigation, obstacle detection, and intelligent decision-making, making them unsuitable for dynamic environments [2]. Additionally, many systems do not support IoT-based real-time data transmission, restricting remote monitoring and control [3]. Therefore, there is a need for a cost-effective, mobile, and autonomous surveillance system that enables real-time monitoring with minimal human intervention.

Another major limitation of existing surveillance systems is their inability to adapt to changing environmental conditions. Fixed camera systems cannot adjust their field of view or reposition themselves based on situational requirements, which leads to blind spots and incomplete monitoring. In large areas such as industrial zones, campuses, and public spaces, this limitation becomes more critical as continuous human monitoring is not always feasible. As a result, there is an increased risk of security breaches going undetected, highlighting the need for more flexible and adaptive surveillance solutions.

Moreover, traditional systems often involve high installation and maintenance costs due to the requirement of multiple cameras, extensive wiring, and dedicated monitoring infrastructure. These systems also consume significant power and require regular supervision to ensure proper functioning. In contrast, modern embedded systems offer compact, energy-efficient, and cost-effective alternatives. However, many existing implementations still lack proper integration of mobility, wireless communication, and real-time processing, which are essential for developing a fully autonomous surveillance system.

Another important challenge is the lack of real-time responsiveness in conventional monitoring systems. Delays in data processing and manual intervention can reduce the effectiveness of surveillance, especially in critical situations. Without intelligent features such as obstacle avoidance, live video streaming, and automated navigation, these systems fail to provide immediate and reliable responses. Therefore, addressing these limitations by integrating IoT, embedded systems, and autonomous robotics is essential for developing an efficient and advanced surveillance solution capable of operating in real-world dynamic environments.

1.3 Objectives and Outcomes

This project aims to design and implement a vision-guided autonomous surveillance rover using embedded systems and Internet of Things (IoT) technologies. The system integrates ESP32 as the main controller and ESP32-CAM for real-time video streaming, enabling continuous monitoring in dynamic environments. Unlike traditional surveillance systems that rely on fixed cameras and manual supervision, this project focuses on developing a mobile and intelligent system capable of autonomous operation with minimal human intervention. The rover is designed to move freely in different environments, capture live visual data, and transmit it wirelessly to a remote user for monitoring and analysis.

The objectives of this project are clearly defined, measurable, and aligned with the identified problem statement. The system is expected to achieve real-time video transmission with low latency, efficient obstacle detection within a range of 2 cm to 400 cm using ultrasonic sensors, and reliable wireless communication using Wi-Fi protocols.

The ESP32 microcontroller is responsible for data acquisition, processing, and communication, ensuring smooth coordination between sensors, actuators, and the camera module. The project is designed to be completed within the given timeframe using cost-effective and energy-efficient components, making it suitable for practical implementation in real-world applications.

Furthermore, the system emphasizes autonomous navigation and intelligent decision-making capabilities through continuous interaction between sensors, control logic, and actuators. The L298 motor driver enables precise control of DC geared motors, allowing accurate directional movement and smooth speed regulation based on real-time sensor inputs. The system dynamically processes obstacle data and adjusts its path accordingly, ensuring safe and efficient navigation without human intervention.

The overall performance is evaluated using parameters such as navigation accuracy, response time, communication stability, and system reliability under various operating conditions. The results indicate that the system maintains consistent performance with stable operation and effective coordination between hardware and software components.

➤ **Primary Objectives:**

- To design and develop a fully autonomous mobile surveillance rover capable of operating without continuous human control.
- To implement real-time video streaming and image capture using ESP32-CAM for live monitoring applications.
- To establish reliable wireless communication using IoT protocols such as Wi-Fi for remote monitoring and control.
- To control motor movement efficiently using L298 motor driver and DC geared motors for precise navigation.
- To develop an embedded system that performs real-time data acquisition, processing, and transmission.

➤ **Secondary Objectives:**

- To minimize human intervention by enabling automated navigation and monitoring.
- To design a cost-effective system using readily available hardware components.
- To ensure low power consumption and efficient energy utilization for longer operation.
- To improve system reliability and performance in real-time conditions.
- To enhance user accessibility through wireless monitoring using mobile or computer interfaces.

➤ **Outcomes:**

- Successful development of a vision-based autonomous surveillance rover with real-time monitoring capability.
- Effective integration of ESP32, ESP32-CAM, ultrasonic sensors, motor driver, and power supply modules.
- Accurate obstacle detection and smooth navigation in different environmental conditions.
- Improved surveillance efficiency compared to traditional fixed monitoring systems.
- A low-cost, scalable, and practical solution suitable for applications such as security monitoring, industrial inspection, and smart surveillance systems.

1.4 Scope and Applications

This project focuses on the design and development of a vision-guided autonomous surveillance rover using embedded systems and IoT technologies. The scope of the project includes the integration of ESP32 microcontroller, ESP32-CAM module, ultrasonic sensor, L298 motor driver, DC geared motors, and a regulated power supply system. The system is developed to perform real-time video streaming, obstacle detection, and autonomous navigation in a controlled environment. The rover captures visual data through the camera module and transmits it wirelessly using Wi-Fi, enabling remote monitoring and control.

The project is limited to prototype-level implementation and testing under controlled conditions. The autonomous navigation is based on ultrasonic sensor data and predefined logic rather than advanced artificial intelligence or machine learning algorithms. The system does not include complex features such as object recognition, facial recognition, or advanced path planning. Additionally, the communication is restricted to Wi-Fi-based IoT connectivity, and cloud integration is implemented only for basic monitoring purposes. Environmental factors such as extreme weather conditions, long-range communication, and large-scale deployment are not considered within the scope of this project.

The system is designed to be completed within the project duration using cost-effective and readily available components. The focus is on achieving reliable performance, real-time responsiveness, and efficient integration of hardware and software modules. The project also emphasizes ease of implementation, low power consumption, and scalability for future enhancements.

In addition to the defined scope, the project emphasizes the development of a modular and flexible system architecture. Each component, including the ESP32 controller, ESP32-CAM module, sensors, and motor driver, is designed to function as an independent unit while maintaining seamless integration with the overall system. This modular approach allows easy troubleshooting, maintenance, and future upgrades. For instance, additional sensors or advanced modules can be incorporated without significantly altering the existing system design, making the project adaptable to evolving technological requirements.

Applications:

- **Security Surveillance:**

The system can be used to monitor restricted or sensitive areas such as offices, industries, warehouses, and residential buildings, providing real-time video surveillance.

- **Industrial Monitoring:**

It can be deployed in industrial environments to inspect machinery, monitor operations, and detect potential hazards without human presence.

- **Smart Home Security:**

The rover can be integrated into smart home systems for remote monitoring and security management.

- **Military and Defence:**

The system can be used for basic reconnaissance and surveillance in controlled or low-risk environments.

- **Search and Rescue Operations:**

It can assist in exploring areas that are difficult or unsafe for humans to access, such as collapsed structures or hazardous zones.

- **Educational and Research Applications:**

The project serves as a practical model for learning embedded systems, IoT, and robotics concepts.

- **Agricultural Monitoring:**

The rover can be used for monitoring crops and farmland conditions in small-scale agricultural environments.

CHAPTER 2 RELATED WORK (or) STATE OF THE ART AND NEED FOR THE PROJECT

2.1 Existing Research, Technologies or Products

The development of autonomous surveillance systems has gained significant attention in recent years due to the increasing demand for intelligent security solutions. Researchers have focused on integrating embedded systems, Internet of Things (IoT), and robotics to design efficient and cost-effective monitoring systems. These systems aim to overcome the limitations of traditional surveillance methods by enabling real-time monitoring, remote accessibility, and autonomous operation.

Research Literature

Several research studies have explored the design and implementation of IoT-based surveillance systems and autonomous robots. IoT-enabled monitoring systems using microcontrollers such as ESP32 have been widely proposed for real-time data transmission and remote access. These systems utilize wireless communication protocols to send sensor data and video streams to cloud platforms for analysis and monitoring [1].

Vision-based surveillance systems have also been developed using camera modules integrated with embedded processors. These systems focus on real-time video streaming and image processing to detect unusual activities. Some advanced approaches incorporate computer vision techniques for object detection and tracking, improving the intelligence of surveillance systems [2].

Autonomous mobile robots with obstacle detection capabilities have been extensively studied. Ultrasonic sensors are commonly used for distance measurement and obstacle avoidance due to their simplicity and cost-effectiveness. These systems enable robots to navigate safely in dynamic environments without human intervention [3].

In addition, research has been conducted on wireless communication and IoT protocols such as MQTT and HTTP for real-time monitoring applications. These technologies ensure efficient data transmission and enable remote control of robotic systems.

Existing Technologies or Commercial Products

Several commercial surveillance solutions are currently available in the market, including fixed CCTV camera systems and smart security devices. Traditional CCTV systems provide continuous monitoring but are limited by fixed installation, restricted coverage, and high installation costs. These systems require multiple cameras and extensive wiring, making them less flexible and scalable.

Modern smart surveillance products include Wi-Fi-enabled cameras, smart doorbells, and mobile monitoring devices. These systems offer features such as real-time video streaming, motion detection, and cloud storage. However, they are still limited by their fixed positioning and lack of mobility, which reduces their effectiveness in large or dynamic environments.

Robotic surveillance systems have also been introduced in industrial and defence applications. These systems are capable of moving in predefined paths and performing monitoring tasks. However, most of these solutions are expensive and complex, making them unsuitable for small-scale or low-cost applications. Additionally, many existing products do not provide full integration of autonomous navigation, real-time video streaming, and IoT-based remote monitoring in a single system.

Relevant Patents

Several patents have been filed in the field of surveillance and autonomous robotic systems. These patents focus on technologies such as mobile surveillance robots, wireless monitoring systems, and intelligent navigation techniques. Some patented systems include robots equipped with cameras and sensors for remote monitoring and security applications.

Patents related to IoT-based surveillance systems highlight the use of wireless communication for real-time data transmission and cloud integration. These systems demonstrate how embedded platforms can be used to enhance surveillance capabilities. However, many patented solutions involve complex architectures and high-cost components, which limit their accessibility for educational and low-cost applications.

2.2 Gaps and Challenges in Existing Approaches

Despite significant advancements in surveillance technologies, several limitations and challenges still exist in current systems. Traditional surveillance methods, such as fixed CCTV systems, are limited by their static nature and restricted field of view. These systems cannot adapt to dynamic environments, resulting in blind spots and incomplete monitoring coverage [1]. Moreover, they rely heavily on continuous human supervision, which increases operational cost and reduces efficiency.

Another major limitation in existing approaches is the lack of mobility and autonomous functionality. Most commercially available surveillance systems are designed for fixed installations and do not support movement or real-time navigation. Although some robotic surveillance systems have been developed, they often rely on predefined paths and lack intelligent decision-making capabilities. This makes them less effective in unpredictable environments where real-time adaptability is essential [2].

IoT-based surveillance systems have improved remote monitoring capabilities; however, they introduce new challenges related to connectivity, latency, and data reliability. Many systems depend on stable internet connections, and any interruption can lead to data loss or delayed responses. Additionally, real-time video streaming requires high bandwidth, which may not always be feasible in low-network conditions. These limitations affect the overall performance and reliability of the system in practical scenarios [3].

Another critical challenge is the cost and complexity of existing advanced surveillance systems. High-end robotic and AI-based surveillance solutions involve expensive hardware components, complex algorithms, and sophisticated infrastructure. This makes them inaccessible for small-scale applications, educational purposes, and low-budget implementations. Furthermore, many systems are not designed with scalability in mind, making it difficult to upgrade or modify them for future requirements [4].

In terms of sensing and navigation, commonly used sensors such as ultrasonic modules have certain limitations. While they are cost-effective and simple to implement, they may fail to detect objects with irregular shapes or soft surfaces that absorb sound waves. Environmental factors such as noise, temperature, and obstacles can also affect sensor accuracy.

Another gap identified in existing systems is the lack of integration between multiple technologies. Many solutions focus either on video surveillance, IoT communication, or robotic movement, but fail to combine all these functionalities into a single efficient system. This lack of integration reduces overall system effectiveness and limits practical usability. A unified system that integrates real-time video streaming, autonomous navigation, and wireless communication is still an area requiring improvement.

Furthermore, most existing systems lack energy efficiency and proper power management. Continuous operation of cameras, sensors, and communication modules consumes significant power, reducing battery life and operational time. This limitation is critical for mobile surveillance systems, where long-duration operation is essential. Efficient power utilization and optimization techniques are required to enhance system performance.

Another significant challenge in existing surveillance approaches is the lack of real-time data processing and decision-making capabilities. Many systems are designed only for data collection and monitoring, without incorporating intelligent processing mechanisms. This results in delayed responses, especially in critical situations where immediate action is required. The absence of real-time analytics and autonomous decision-making reduces the effectiveness of these systems in dynamic and unpredictable environments.

Scalability is also a major concern in current surveillance solutions. Most systems are designed for specific applications and cannot be easily expanded or modified to meet changing requirements. Adding new components or upgrading system capabilities often requires significant redesign and additional cost. This limitation makes it difficult to deploy such systems in large-scale environments where flexibility and scalability are essential.

Interoperability between different hardware and software components is another challenge faced by existing systems. Many surveillance solutions use proprietary technologies and protocols, which restrict compatibility with other devices and platforms. This lack of standardization limits the ability to integrate multiple systems into a unified solution, thereby reducing overall efficiency and usability.

2.3 Need for the Proposed Project

The rapid growth of urbanization and industrialization has significantly increased the demand for efficient and intelligent surveillance systems. Traditional surveillance methods, such as fixed CCTV cameras, are no longer sufficient to meet modern security requirements due to their limited coverage, lack of mobility, and dependence on continuous human monitoring [1]. These limitations create critical gaps in monitoring large and dynamic environments, where real-time adaptability and autonomous operation are essential.

As discussed in the previous section, existing surveillance systems face several challenges, including lack of autonomous navigation, limited integration of technologies, high implementation costs, and dependency on stable network infrastructure. While IoT-based systems have improved remote monitoring capabilities, they still suffer from issues such as latency, data reliability, and restricted scalability [2]. These shortcomings highlight the need for a system that not only provides real-time monitoring but also ensures flexibility, mobility, and efficient performance in diverse environments.

Another important factor driving the need for this project is the increasing requirement for cost-effective and easily deployable surveillance solutions. Many advanced surveillance systems and robotic platforms available in the market are expensive and complex, making them unsuitable for small-scale applications, educational purposes, and low-budget implementations. There is a clear need for a system that utilizes affordable components while maintaining acceptable performance and reliability [3].

Furthermore, the absence of fully integrated systems combining vision-based monitoring, autonomous navigation, and IoT communication creates a significant research gap. Most existing solutions focus on individual functionalities rather than providing a unified system. This lack of integration reduces the overall effectiveness and practical usability of surveillance systems. Therefore, a solution that combines these technologies into a single platform is essential to enhance system capability and efficiency [4].

CHAPTER-3 REQUIREMENTS AND CONSTRAINTS

3.1 Requirements

The successful implementation of the vision-guided autonomous surveillance rover requires a well-defined set of functional and performance requirements. These requirements play a crucial role in ensuring that the system operates efficiently, reliably, and meets the objectives outlined in the earlier stages of the project. Since the system integrates multiple hardware and software components such as the ESP32 microcontroller, ESP32-CAM module, ultrasonic sensor, motor driver, and wireless communication modules, it is essential to clearly define how each component contributes to the overall functionality. To achieve better clarity and organization, the requirements are categorized into functional requirements and performance requirements, which together define the system behaviour and expected outcomes.

Functional requirements describe what the system is expected to do during operation. The rover must be capable of autonomous navigation by continuously monitoring its surroundings and detecting obstacles using the ultrasonic sensor. Based on the sensor input, the system should make real-time decisions to stop, turn, or change direction to avoid collisions. In addition to navigation, the system must support real-time image capture using the ESP32-CAM module and transmit the visual data over a wireless network for remote monitoring. The ESP32 microcontroller should act as the central controller, coordinating all operations such as sensor data processing, motor control, and communication. The system should also provide basic user interaction features, such as displaying status information through an LCD or monitoring interface. These functional requirements ensure that the system performs all intended tasks effectively.

Performance requirements, on the other hand, define how well the system should perform its functions under various conditions. The system must operate with minimal delay, ensuring quick response to sensor inputs for effective obstacle avoidance. The accuracy of the ultrasonic sensor should be maintained within its specified range to ensure reliable distance measurement. Overall, the definition of functional and performance requirements provides a strong foundation for system design and implementation.

3.1.1 Functional Requirements

Functional requirements define the essential operations that the system must perform to achieve its intended purpose. These requirements describe how various hardware and software components interact and work together to deliver the desired functionality. In the case of the vision-guided autonomous surveillance rover, functional requirements ensure that the system is capable of performing tasks such as monitoring, navigation, communication, and user interaction in an efficient and coordinated manner.

One of the primary functional requirements of the system is the ability to capture and transmit real-time visual data. The ESP32-CAM module is responsible for capturing images or video streams from the rover's surroundings. This data is then transmitted wirelessly using the built-in Wi-Fi capability of the ESP32, enabling remote users to monitor the environment in real time. This feature is essential for surveillance applications, as it allows continuous observation without the need for physical presence. The system must ensure that image capture and transmission occur seamlessly, with minimal delay and sufficient clarity for effective monitoring.

The ESP32 microcontroller acts as the central processing unit of the system and plays a critical role in coordinating all operations. It continuously collects data from input devices such as the ultrasonic sensor and processes this information to make real-time decisions. Based on the processed data, the ESP32 generates control signals to manage motor operations and communication processes. Its ability to handle multiple tasks simultaneously ensures smooth interaction between sensing, processing, and actuation components.

Another key functional requirement is autonomous movement. The rover must be capable of moving in multiple directions, including forward, backward, left, and right, using DC geared motors controlled by the L298 motor driver. The system is programmed with logic that allows it to navigate independently without continuous human intervention. Obstacle detection is an integral part of this functionality, achieved using the ultrasonic sensor. The sensor continuously measures the distance to nearby objects, and when an obstacle is detected within a predefined range, the system automatically takes corrective action such as stopping or changing direction to avoid collisions. This ensures safe and efficient navigation in dynamic environments.

3.1.2 Performance Requirements

Performance requirements define how well the system should operate under various conditions and are essential to evaluate the efficiency, reliability, and overall effectiveness of the vision-guided autonomous surveillance rover. These requirements focus on key parameters such as speed, accuracy, response time, reliability, and power efficiency, ensuring that the system performs consistently in real-time environments.

One of the most important performance requirements is real-time video streaming. The system should provide continuous video transmission using the ESP32-CAM module with minimal latency to ensure effective monitoring. The delay in video transmission must be low enough to allow users to observe live conditions and respond promptly without significant lag. The ESP32-CAM should capture and transmit images efficiently, maintaining acceptable frame rates and minimizing frame drops to ensure smooth visual output. This is critical for surveillance applications where timely information is essential.

Another key aspect is the accuracy and responsiveness of the ultrasonic sensor used for obstacle detection. The sensor should reliably measure distances within its operational range of 2 cm to 400 cm, maintaining consistent accuracy under different environmental conditions. The response time of the sensor must be fast enough to allow the system to detect obstacles instantly and take appropriate action. Quick response to sensor inputs ensures that the rover can avoid collisions and navigate safely in dynamic environments.

The motion performance of the rover is also an important requirement. The system should exhibit smooth and stable movement with controlled speed and direction. The L298 motor driver, in combination with DC geared motors, should provide sufficient torque to enable movement across various surfaces while maintaining balance and stability. The system should also demonstrate consistent performance without jerks or instability, even during direction changes or obstacle avoidance manoeuvres. Efficient power utilization by the motor driver is essential to ensure prolonged operation without excessive energy consumption.

Power efficiency is another crucial performance requirement. The system should be designed to consume minimal power while maintaining optimal functionality.

3.2 Engineering Standards and Codes

The design and implementation of the vision-guided autonomous surveillance rover must comply with various engineering standards and codes to ensure safety, reliability, interoperability, and performance. These standards guide the development of both hardware and software components, ensuring that the system meets industry best practices and operates efficiently under defined conditions.

□ Hardware Standards

➤ *IEEE Standards*

IEEE standards play a crucial role in communication and embedded system design. The IEEE 802.11 standard defines wireless communication protocols (Wi-Fi), which are used in the ESP32 and ESP32-CAM modules for real-time data transmission and remote monitoring [1]. Compliance with this standard ensures reliable wireless connectivity, interoperability with other devices, and efficient data transfer.

➤ *IEC (International Electrotechnical Commission) Standards*

IEC standards ensure electrical safety and performance of electronic systems. Standards such as IEC 60335 (safety of electrical appliances) and IEC 61000 (electromagnetic compatibility) are relevant to this project. These standards influence the design by ensuring that the system operates safely without causing electromagnetic interference to other devices and can withstand electrical disturbances [2].

➤ *ISO Standards (Hardware & System Safety)*

ISO standards such as ISO 12100 (safety of machinery) provide guidelines for designing safe mechanical systems. In this project, these standards influence the design of the robotic rover, ensuring safe movement, proper mechanical structure, and reduced risk of accidents during operation [3].

□ **Software and Communication Standards**

➤ *ISO/IEC 25010 (Software Quality Model)*

ISO/IEC 25010 defines software quality characteristics such as reliability, performance efficiency, usability, and maintainability. This standard influences the design of the embedded software running on the ESP32, ensuring that the system is stable, efficient, and easy to maintain [4].

➤ *IoT Communication Protocol Standards*

Protocols such as MQTT and HTTP follow standardized communication models for IoT systems. These standards ensure efficient and secure data transmission between devices and cloud platforms. The use of standardized protocols improves interoperability, scalability, and real-time performance of the system.

➤ *Data Security and Privacy Standards*

Although this project is a prototype, basic data security principles must be followed. Standards such as general data protection and secure communication practices influence the design by encouraging the use of encrypted communication (e.g., HTTPS) and secure authentication methods. This ensures protection against unauthorized access and data breaches.

➤ *Coding Standards*

Standard coding practices are followed to ensure readability, maintainability, and error-free implementation. Structured programming, proper variable naming, modular design, and debugging techniques are applied while developing the embedded software. This improves system reliability and simplifies future modifications.

□ **Environmental and Safety Considerations**

The system must adhere to general safety guidelines to ensure safe operation in real-world environments. Proper insulation, secure wiring, and stable mounting of components are necessary to prevent electrical hazards and mechanical failures. Additionally, the system should be designed to operate within specified temperature and environmental conditions to ensure consistent performance.

3.3 Design Constraints

The design and implementation of the vision-guided autonomous surveillance rover are influenced by several practical constraints. These constraints must be carefully considered to ensure that the system is feasible, reliable, and suitable for real-world applications. The following constraints outline the limitations and factors affecting the project.

□ **Economic Constraints**

The project is designed to be cost-effective by using affordable and widely available components such as ESP32, ESP32-CAM, ultrasonic sensors, and L298 motor driver. Budget limitations restrict the use of high-end sensors, advanced processors, or expensive robotic platforms. As a result, the system focuses on achieving optimal performance within a low-cost framework. This constraint directly influences component selection, system design, and overall functionality.

□ **Environmental Constraints**

The system is designed to operate in normal indoor and controlled outdoor environments. It is not intended for extreme environmental conditions such as heavy rain, high humidity, or very high/low temperatures. Environmental factors such as dust, noise, and obstacles may affect sensor accuracy and system performance. Therefore, the project scope is limited to environments where stable operation can be ensured.

□ **Sustainability Constraints**

The project considers basic sustainability aspects by using low-power components such as ESP32 and efficient DC motors. However, the system relies on battery power, which requires periodic charging and replacement. Advanced energy optimization techniques and renewable energy sources are not implemented in this prototype.

□ **Manufacturability Constraints**

The system is developed as a prototype using readily available components and standard assembly techniques. No complex manufacturing processes or industrial fabrication methods are involved. The design is simple and modular, allowing easy assembly and maintenance. Therefore, large-scale manufacturing constraints are not directly applicable but are considered for future scalability.

□ **Ethical Constraints**

The project involves surveillance and monitoring, which may raise privacy concerns if used in real-world applications. Ethical considerations include ensuring that the system is used only for authorized monitoring and does not violate individual privacy. Proper usage guidelines and access control mechanisms must be followed. Although this is a prototype, ethical implications are acknowledged and considered.

3.3.1 Economic Constraints

Cost is a key constraint, requiring low-cost components and minimal overall budget. The system uses affordable hardware and open-source software tools. This ensures cost-effectiveness while maintaining basic performance.

- **Software:** The project uses open-source tools such as Arduino IDE, eliminating licensing costs. Development is carried out using freely available libraries, ensuring minimal software expenditure.

- **Hardware:** The total hardware cost is maintained within an affordable range (approximately ₹5,000–₹10,000) by selecting low-cost components such as ESP32, ESP32-CAM, L298 motor driver, ultrasonic sensor, and DC gear motors. Preference is given to locally available components to reduce procurement and maintenance costs.

- **Hybrid:** Budget constraints influence both hardware and software decisions, ensuring a balance between cost and performance. The system is designed to be economical while delivering acceptable functionality, making it suitable for academic use and small-scale real-world applications.

In addition to the defined budget limits, cost optimization is further achieved by selecting **multi-functional components** such as ESP32, which combines processing capability and built-in Wi-Fi communication.

The project also avoids the use of **high-cost sensors and advanced processing units**, such as LiDAR or high-resolution industrial cameras, which are typically used in commercial surveillance systems. Instead, simpler and cost-effective alternatives like ultrasonic sensors are used for obstacle detection. Although this may slightly limit precision, it significantly reduces the overall system cost while still meeting the basic functional requirements.

Maintenance and replacement costs are also considered as part of the economic constraint. The use of commonly available and standardized components ensures that faulty parts can be easily replaced without incurring high expenses. This makes the system more sustainable and practical for long-term usage, especially in educational environments.

Furthermore, the project minimizes operational costs by relying on **local data processing and direct Wi-Fi communication**, avoiding the need for continuous cloud storage or subscription-based services. This ensures that the system can function without recurring expenses, making it more economical over time.

Finally, the economic constraint influences the overall system design by encouraging a **balance between cost and performance**. While high-end features are limited due to budget restrictions, the system is optimized to deliver reliable performance within the available resources. This approach ensures that the project remains practical, scalable, and suitable for further development without significant financial burden.

3.3.2 *Environmental and Sustainability Constraints*

Environmental conditions are important considerations in the design of the proposed surveillance rover. The system is intended to operate in typical indoor and semi-outdoor environments, where factors such as dust, minor temperature variations, and physical obstacles may be present.

- **Software:** The system software is designed to be efficient by minimizing processing load and avoiding unnecessary data transmission. Optimized code execution reduces CPU usage, which in turn lowers energy consumption and enables smooth operation on low-power embedded platforms such as ESP32.

- **Hardware:** Components are carefully selected to ensure low power consumption and energy efficiency, with total system power maintained below ~5W.

- **Hybrid:** The combination of optimized software and low-power hardware reduces overall system energy consumption and battery drain. This integrated approach enhances efficiency, supports longer operation time, and promotes sustainability by minimizing both power usage and e-waste generation.

In addition to energy efficiency, the project also considers **recyclability and responsible material usage**. Wherever possible, components such as PCB boards, wiring, and structural parts are selected based on their ability to be reused or recycled. The use of RoHS-compliant electronic components ensures that hazardous substances such as lead and mercury are minimized, thereby reducing environmental impact and promoting eco-friendly design practices.

Power consumption is carefully monitored to ensure that the total system operates within a **low power range (approximately <5W)**. Efficient power regulation techniques are implemented to avoid unnecessary energy loss during operation. The use of rechargeable batteries further supports sustainability by reducing dependency on disposable power sources and minimizing long-term environmental waste.

From a software perspective, additional optimization is achieved by reducing **redundant data transmission and processing overhead**. The system avoids continuous high-load operations by transmitting only essential data, such as real-time video when required and sensor readings at controlled intervals. This approach reduces network load, processing demand, and overall energy consumption, making the system more efficient and environmentally friendly.

The project also minimizes **electronic waste (e-waste)** by adopting a modular design approach. Individual components such as sensors, microcontroller units, and motor drivers can be replaced or upgraded independently without discarding the entire system. This significantly extends the lifecycle of the system and reduces unnecessary waste generation.

Thus, the proposed system promotes sustainability by enabling efficient resource utilization and reducing environmental impact. The modular design not only enhances flexibility and maintainability but also supports long-term usability through easy upgrades and replacements. Overall, this approach ensures that the system remains cost-effective, environmentally responsible, and adaptable to future technological advancements.

3.3.3 *Manufacturability Constraints*

Manufacturability plays a critical role in ensuring that the proposed vision-guided autonomous surveillance rover can be easily fabricated, assembled, and replicated using standard resources. The system is designed as a prototype that can be developed within a laboratory or academic environment without requiring advanced industrial manufacturing processes.

One of the key constraints is the **availability of components in local markets**. The project relies on commonly used modules such as ESP32, ESP32-CAM, ultrasonic sensors, L298 motor driver, and DC geared motors, which are widely available from local electronics suppliers. This reduces procurement delays and avoids dependency on imported or specialized components. The design avoids rare or custom-built parts, ensuring that replacement and maintenance are straightforward and cost-effective.

Another important constraint is the **simplicity of assembly and fabrication**. The system is designed to be assembled using basic tools such as soldering irons, screwdrivers, and breadboards. Complex fabrication processes such as custom PCB manufacturing, CNC machining, or advanced 3D printing are avoided. Instead, the project uses ready-made development boards and modular components that can be easily connected using jumper wires and standard connectors. This ensures that the system can be built by students or beginners without requiring specialized technical expertise.

Mechanical design is also influenced by manufacturability constraints. The rover chassis must be simple, lightweight, and easy to assemble using readily available materials such as plastic or metal frames. The mounting of motors, sensors, and camera modules should be straightforward, with minimal alignment complexity. The design should allow easy access to components for maintenance and upgrades, ensuring long-term usability.

Additionally, standardization of components and interfaces is considered to improve manufacturability and system integration. This approach simplifies assembly, testing, and maintenance, while also enabling easy replacement or upgrading of components. Furthermore, it supports scalability and reproducibility, making the system easier to replicate and deploy in practical applications.

•Software:

The software must be designed to work seamlessly with widely available hardware platforms such as ESP32 and must be developed using open-source environments like Arduino IDE. The code should be modular, well-structured, and easy to understand, allowing developers to modify or extend functionality without difficulty. Firmware uploading and debugging should be possible using standard USB interfaces without requiring specialized programming hardware. The software should also support cross-platform compatibility, enabling development and monitoring through commonly available systems such as Windows, Linux, or Android devices.

•Hardware:

The hardware must be designed for easy assembly using commonly available electronic components and tools. The use of pre-built modules eliminates the need for complex circuit design and PCB fabrication. All components should be easily mountable on a standard chassis using screws, brackets, or adhesive supports. Wiring should be simple and well-organized to avoid short circuits and ensure safe operation. The design must also consider durability, ensuring that components remain securely connected during movement. Furthermore, the selection of components must ensure that replacements are readily available, reducing downtime in case of failure. The system should also be designed in such a way that it can be disassembled and reassembled without damaging any components.

•Hybrid:

The integration of software and hardware must support a smooth and repeatable development process. The ESP32 microcontroller acts as a bridge between physical components and embedded software, enabling seamless communication and control. The modular architecture ensures that each component can be independently tested before full system integration. This approach simplifies debugging and reduces the chances of system failure. Additionally, the system is designed to allow easy upgrades, such as adding new sensors or improving communication modules, without requiring major redesign. The overall design ensures that the project can be replicated by other students or developers using similar resources, making it highly practical and scalable.

The overall architecture promotes ease of replication, making it practical for other students or developers to build, modify, and extend the system using commonly available resources. This results in a robust, maintainable, and future-ready design.

3.3.4 *Ethical and Health & Safety Constraints*

The proposed vision-guided autonomous surveillance rover involves monitoring and data transmission, which introduces important ethical and safety considerations. Although the system is developed as a prototype, it must still ensure responsible usage, user safety, and compliance with general ethical guidelines. The system is designed to operate in a controlled environment with minimal risk; however, potential concerns related to privacy, data handling, and physical safety are carefully addressed.

□ **Ethical Considerations**

The primary ethical concern in this project is **user privacy and responsible surveillance**. Since the system captures real-time video using the ESP32-CAM module, it must be ensured that monitoring is performed only in authorized areas and for legitimate purposes. Unauthorized surveillance or misuse of recorded data must be strictly avoided. The system should not store or share sensitive visual data without proper permission.

Informed usage is another important ethical aspect. Users operating the system must be aware of its functionality and limitations. If deployed in real-world environments, appropriate consent should be obtained from individuals in the monitored area. Although this is a prototype, the design acknowledges the importance of ethical surveillance practices. The system does not implement advanced AI algorithms; therefore, issues related to **algorithmic bias** are minimal.

□ **Health and Safety Considerations**

The system is designed to operate under **low-voltage conditions (<12V)**, ensuring minimal electrical risk to users. Proper insulation, secure wiring, and stable connections are maintained to prevent short circuits, electric shocks, or component damage. The use of regulated power supply components ensures stable voltage levels across all modules.

Mechanical safety is also considered in the rover design. The system avoids sharp edges and unstable structures that could cause injury. Motors and moving parts are securely mounted to prevent accidental detachment during operation. The speed of the rover is kept within safe limits to avoid collisions or damage to surroundings.

Thermal safety is another important factor. Components such as voltage regulators and motor drivers may generate heat during operation.

□ **Compliance with Standards**

Although the project is a prototype, it aligns with general engineering safety and ethical standards:

- **IEC Electrical Safety Standards** ensure safe handling of low-voltage electronic systems.
- **ISO Safety Guidelines** influence safe mechanical design and operation.
- **Basic Data Privacy Principles (similar to GDPR concepts)** guide responsible handling of transmitted data.

These standards help ensure that the system is safe, reliable, and ethically acceptable for its intended use.

□ **Software Perspective**

From a software perspective, the system is designed to ensure **secure and efficient data handling**. Only necessary data is processed and transmitted, reducing exposure to potential security risks. The software avoids unnecessary storage of sensitive information and ensures controlled access to the system.

The system does not provide automated decision-making that could harm users, and therefore the risk of harmful recommendations is minimal. However, proper coding practices and testing are followed to avoid system errors that could lead to unsafe behavior.

□ **Hardware Perspective**

From a hardware perspective, the system ensures **safe physical operation**. All electronic components operate within safe voltage and current limits. The design avoids the use of hazardous materials and ensures proper mounting of all components. The rover structure is stable and designed to prevent accidental tipping or damage.

No direct human contact components (such as biomedical sensors) are involved, so biocompatibility constraints are not applicable to this project.

□ **Hybrid Perspective**

The integration of hardware and software ensures overall system safety and ethical compliance. The hardware provides a safe physical platform, while the software ensures controlled operation and secure data handling. Together, they create a system that minimizes risks related to both physical hazards and ethical concerns.

3.3.5 Social Constraints

The proposed vision-guided autonomous surveillance rover must consider various social factors such as usability, accessibility, user acceptance, and cultural sensitivity. Since the system is intended for surveillance purposes, it is important to ensure that its operation does not create discomfort, fear, or misunderstanding among users. The design focuses on making the system simple, user-friendly, and socially acceptable for use in environments such as homes, industries, and educational institutions.

▣ *Software Perspective*

From a software perspective, the system must ensure **ease of use and accessibility**. The user interface used for monitoring (such as mobile or web-based interface) should be simple and intuitive, allowing users with basic technical knowledge to operate the system. The system should avoid complex commands and provide clear visual outputs, such as live video streams, that are easy to interpret. Although the current prototype may use a basic interface, future improvements can include **multilingual support** to make the system accessible to users from different linguistic backgrounds. Additionally, the software should avoid generating confusing or misleading outputs. The system does not include sensitive data processing; therefore, risks related to stigmatization are minimal. However, clear and neutral representation of information is maintained.

▣ *Hardware Perspective*

From a hardware perspective, the rover must be designed in a way that is **socially acceptable and non-intrusive**. The size and structure of the rover should be compact and not intimidating to users. The movement of the rover should be smooth and controlled to avoid causing disturbance in the surrounding environment. The system does not involve any invasive procedures or direct human interaction, which reduces social concerns. The design ensures that the rover can operate safely in shared environments without causing inconvenience. The physical appearance and operation of the device are kept simple and functional, avoiding any design elements that could create discomfort or negative perception.

▣ *Hybrid Perspective*

The integration of hardware and software ensures that the overall system is **user-friendly and socially adaptable**. The hardware provides a compact and safe platform, while the software ensures that the system is easy to monitor and control. Together, they create a solution that can be comfortably used in different social environments without requiring specialized knowledge.

The system is designed to support **remote monitoring**, reducing the need for continuous human presence and making it convenient for users. At the same time, it ensures that surveillance is performed responsibly, without violating social norms or creating unnecessary concern among individuals. In addition to usability and accessibility, the system must also consider **user trust and acceptance**, which are critical factors in surveillance-related applications. People may feel uncomfortable or concerned when being monitored by a moving robotic device. Therefore, the system must be used in a transparent manner, where its purpose is clearly communicated to users. Proper awareness about the system's functionality helps in reducing fear and improving acceptance in social environments.

Another important consideration is the **impact of the system on daily activities**. The rover should operate in a way that does not disturb normal human activities in its surroundings. Its movement should be smooth, controlled, and predictable, avoiding sudden or aggressive actions that could startle individuals. The noise generated by motors and components should be minimal to ensure a comfortable environment, especially in indoor settings such as homes, offices, and classrooms. Accessibility is further enhanced by ensuring that the system can be used by individuals with varying levels of technical knowledge. The control and monitoring interface should not require advanced training, making it suitable for a wide range of users. Future improvements may include voice control or simplified mobile interfaces to further enhance accessibility for elderly or non-technical users.

Furthermore, the system should promote **positive social impact** by enhancing safety and security rather than creating fear or misuse. The project also considers **user interaction and feedback** as part of social design. Although the current prototype has limited interaction features, future enhancements can include alert systems, status indicators, or simple notifications that inform users about system activity. This improves transparency and helps users stay informed about the system's operation.

3.3.6 *Political/Regulatory Constraints*

The proposed vision-guided autonomous surveillance rover must consider applicable political and regulatory factors, especially when deployed in real-world environments. Although the current project is developed as an academic prototype, surveillance systems are generally subject to government policies, data protection laws, and safety regulations. Therefore, it is important to ensure that the system aligns with basic legal and regulatory requirements to enable safe and compliant usage.

Software Perspective

From a software perspective, the system must comply with established data protection and privacy regulations to ensure ethical and secure operation. Since the system involves real-time video monitoring, it is important to prevent unauthorized data collection, storage, or sharing. Even though the current prototype does not store data permanently, appropriate safeguards must be considered for future enhancements. Any extended implementation should follow established data protection principles similar to regulations such as GDPR (General Data Protection Regulation) or relevant national data privacy laws. This includes ensuring user consent, limiting data usage to intended purposes, and avoiding unnecessary data retention. The software design must also ensure that only authorized and properly licensed tools, APIs, and libraries are used during development. The use of open-source or legally compliant software components helps maintain transparency and avoids legal complications. Additionally, the system should adhere to local internet usage policies and regulatory guidelines related to wireless communication. This ensures that the system operates within permissible limits and does not interfere with restricted frequencies or violate communication standards.

Furthermore, secure communication practices are essential to protect the system from unauthorized access and potential cyber threats. When the system is connected to a network, appropriate security measures such as encrypted communication protocols, secure authentication mechanisms, and controlled access should be considered. In addition to hardware and software considerations, proper system validation and compliance checks should be carried out throughout the development process.

❏ *Hardware Perspective*

From a hardware perspective, the system is designed to comply with basic electronic and safety regulations applicable to low-voltage embedded systems. Since the rover operates under low voltage conditions (typically less than 12V), it falls within safe operating limits and minimizes the risk of electrical hazards such as shocks or short circuits. This makes the system suitable for academic and prototype-level applications without requiring strict industrial certifications. However, despite operating at low voltage, proper design and safety practices are strictly followed to ensure reliable and secure usage.

All hardware components are assembled with careful attention to insulation, wiring quality, and secure connections. Loose connections and improper wiring can lead to system instability or component damage, hence standardized assembly techniques are adopted. The use of a regulated power supply ensures consistent voltage levels are maintained across all modules, protecting sensitive components such as the ESP32 and sensors from voltage fluctuations. Proper grounding and circuit protection measures are also considered to enhance safety and prevent unexpected failures during operation. The selection of components is based on availability, cost-effectiveness, and compliance with general usage regulations. The project avoids the use of restricted, hazardous, or export-controlled components, ensuring that all hardware elements are legally permitted and easily accessible in local markets. Components such as ESP32, ESP32-CAM, ultrasonic sensors, and motor drivers are widely used in educational and industrial prototyping environments, making the system easy to replicate and maintain. This also reduces dependency on specialized or imported components, improving overall feasibility.

The system is intended for general surveillance and monitoring applications and does not fall under specialized regulatory domains such as medical, aerospace, or defense-grade equipment. Therefore, strict regulatory approvals or certifications are not required at the prototype level. Furthermore, electromagnetic interference (EMI) and signal stability are considered during hardware design to ensure that wireless communication modules function effectively without causing interference to other nearby devices. The ESP32's built-in Wi-Fi operates within standard frequency ranges, adhering to common communication norms. This ensures that the system can function smoothly in typical environments without violating communication guidelines.

❏ *Hybrid Perspective*

The integration of hardware and software in the proposed system must ensure compliance with both safety standards and data protection regulations to achieve secure and reliable operation. While the hardware components are responsible for safe physical functioning of the rover, the software ensures responsible handling of data and secure communication. This combined approach is essential, especially when the system is intended for deployment in real-world environments such as public or sensitive areas. Proper coordination between hardware safety measures and software-level security practices ensures that the system operates within acceptable regulatory and ethical boundaries. In real-world applications, particularly in public environments, additional approvals and permissions may be required based on local government policies and institutional guidelines. These may include authorization for surveillance activities, compliance with privacy laws, and adherence to specific usage restrictions. The system must be deployed only in authorized locations to prevent misuse and avoid legal complications. Ensuring proper permissions before deployment helps maintain transparency and accountability, especially when the system is used for monitoring or security purposes.

Furthermore, the system must consider region-specific regulations related to surveillance and electronic communication. Different countries and states have varying rules governing the use of cameras and monitoring devices in public and private spaces.

Another important aspect of regulatory compliance involves wireless communication. The ESP32 module operates using Wi-Fi frequencies that must conform to government-approved spectrum regulations. The system should function within standard frequency bands and transmission limits to avoid interference with other communication systems. Adhering to these regulations ensures reliable communication performance while preventing unauthorized use of wireless channels. It also supports compatibility with existing network infrastructures.

In addition, the system is designed to follow basic cybersecurity principles by avoiding the use of restricted encryption techniques or unauthorized network access methods. Any future enhancements involving cloud connectivity, remote servers, or data storage must comply with national cybersecurity frameworks and data protection laws.

CHAPTER-4 DESIGN AND IMPLEMENTATION

4.1 Concept Development and Alternatives

During the initial phase of the project, multiple design concepts and implementation approaches were carefully analysed to develop an efficient, reliable, and intelligent surveillance rover. A systematic evaluation process was carried out to explore various alternatives in terms of hardware selection, sensor integration, communication methods, and mobility mechanisms. This phase played a crucial role in identifying the most suitable design that balances performance, cost, and feasibility within the given project constraints. Several microcontroller options such as Arduino-based boards, Raspberry Pi, and ESP32 were considered during the design stage. Each option was evaluated based on processing capability, power consumption, ease of programming, and support for wireless communication. The ESP32 was selected due to its dual-core processing capability, built-in Wi-Fi and Bluetooth support, and cost-effectiveness, making it highly suitable for real-time control and communication tasks. Similarly, different sensor configurations were analysed, including infrared sensors and ultrasonic sensors for obstacle detection. The ultrasonic sensor was chosen for its better accuracy, wider detection range, and reliability under varying environmental conditions.

The evaluation process involved a detailed comparison of each alternative, considering factors such as system complexity, scalability, energy efficiency, and ease of integration. Simpler designs were preferred to reduce implementation difficulty and ensure ease of maintenance, while still achieving the desired functionality. The selected components and design approach offer a good balance between performance and cost, making the system practical for both academic and real-world applications.

Additionally, feasibility analysis was conducted to ensure that all selected components are readily available in local markets and can be easily assembled using standard tools and techniques. This avoids dependency on specialized equipment or imported components, thereby reducing cost and development time. The design also emphasizes modularity, allowing future upgrades such as advanced sensors or improved communication modules without significant changes to the existing system.

4.1.1 *Concept A – Manual Controlled Surveillance Rover*

In this approach, the rover is controlled manually using a remote device or mobile application. The user directly controls the movement of the rover and monitors the environment through a camera feed.

Strengths:

- Simple design and implementation
- Easy control and operation
- Low processing requirement

Weakness:

- Requires continuous human intervention
- Not suitable for autonomous operation
- Limited efficiency in large-scale monitoring

4.1.2 *Concept B – Sensor-Based Autonomous Rover*

This concept involves an autonomous rover that uses sensors such as ultrasonic sensors for navigation and obstacle avoidance. The rover operates based on predefined logic without human intervention.

Strengths:

- Autonomous movement
- Improved efficiency compared to manual systems
- Safe navigation using obstacle detection

Weakness:

- Limited intelligence (no visual monitoring)
- Cannot perform surveillance effectively

4.1.3 *Concept C – Vision-Guided Autonomous Surveillance Rover*

This concept integrates both vision and sensor technologies. The rover uses an ESP32-CAM module for visual monitoring and ultrasonic sensors for obstacle detection.

Strengths:

- Combines vision and autonomous navigation
- Real-time monitoring capability

Weakness:

- Slightly higher complexity
- Requires proper integration of components

4.1.4 Evaluation Criteria

The concepts were evaluated based on the following criteria:

- ❖ Level of automation
- ❖ Cost of implementation
- ❖ System complexity
- ❖ Reliability and performance
- ❖ Suitability for surveillance applications

Based on these criteria, Concept C was selected as the final design due to its superior performance and ability to meet project objectives.

4.2 Final Design and Justification

The final design selected for the project is the Vision-Guided Autonomous Surveillance Rover, which integrates key components such as the ESP32 microcontroller, ESP32-CAM module, ultrasonic sensors, motor driver, and DC geared motors. This design was chosen after a detailed evaluation of multiple alternatives, ensuring that it meets the required functional and performance criteria while remaining cost-effective and easy to implement.

Table 4.1 presents a comparative analysis of the proposed design alternatives based on important evaluation parameters such as cost, feasibility, performance, and functionality. Each design option was assessed systematically to identify its strengths and limitations. Based on this comparison, the selected design achieved the highest overall rating by providing an optimal balance between affordability, system efficiency, and implementation simplicity. It effectively satisfies the functional requirements and design constraints outlined in Chapter 3, while also ensuring reliability and scalability for future improvements.

The chosen design enables key functionalities such as autonomous navigation, real-time image capture, and wireless communication. The integration of the ESP32-CAM module enhances the system's vision capabilities, allowing continuous monitoring and real-time data transmission. At the same time, the ultrasonic sensor ensures accurate obstacle detection, enabling safe and efficient movement of the rover in dynamic environments. This combination of vision-based monitoring and sensor-based navigation significantly improves the overall effectiveness of the system.

Table 4.1 Comparison of Alternative Solutions

Criteria	Concept A (Manual controlled rover)	Concept B (Sensor-based autonomous rover)	Concept C (Vision-Guided autonomous rover)
Cost	✓ Low	✗ Medium	✗ High
Feasibility	✓ Easy to deploy	✓ Moderate	✗ Complex hardware build
Accuracy	✗ Low (manual dependency)	✓ High	✓ Moderate
Compliance with Standards	✓ Basic	✓ moderate	✓ Strong
Overall Assessment	Moderate	Good	Best choice

Table 4.2 Trade-off Analysis of Design Alternatives

Criteria	Weight	Concept A (Basic monitoring system)	Concept B (Speech + Text AI)
Cost (budget limits)	0.25	4 (low cost, Fewer components)	3 (camera module increases cost)
Feasibility (time/resources)	0.20	4 (easy to implement)	4 (requires integration of camera module)
Accuracy (meets performance requirement)	0.30	2 (no visual monitoring, Limited accuracy)	4 (Real-Time image monitoring improves accuracy)
Compliance with Standards (safety, System reliability)	0.25	3(basic compliance)	4(Enhanced monitoring and safety)
Weighted Score	1.00	3.10	3.50

4.3 System Architecture

The system architecture defines the overall structure and interaction between different components of the system. The proposed system follows an input-process-output model.

Input: Camera module (ESP32-CAM), Ultrasonic sensor

Processing: ESP32 microcontroller

Output: Motor driver (L298), DC motors, LCD display

The ESP32 acts as the central unit that collects data from sensors, processes it, and controls the output devices accordingly. Wireless communication enables real-time monitoring and data transmission.

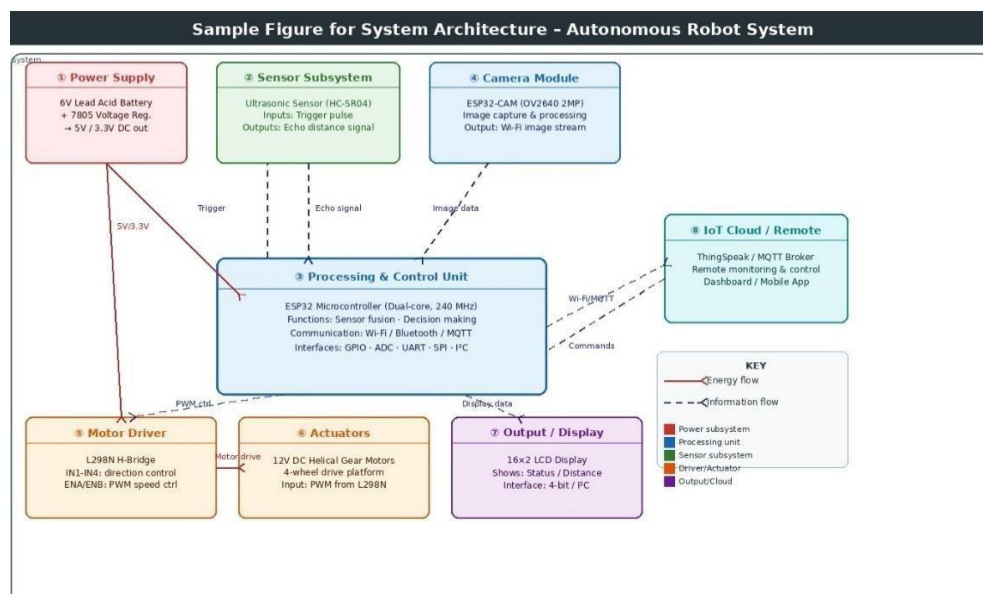


Figure 4.1 Sample Figure for System Architecture

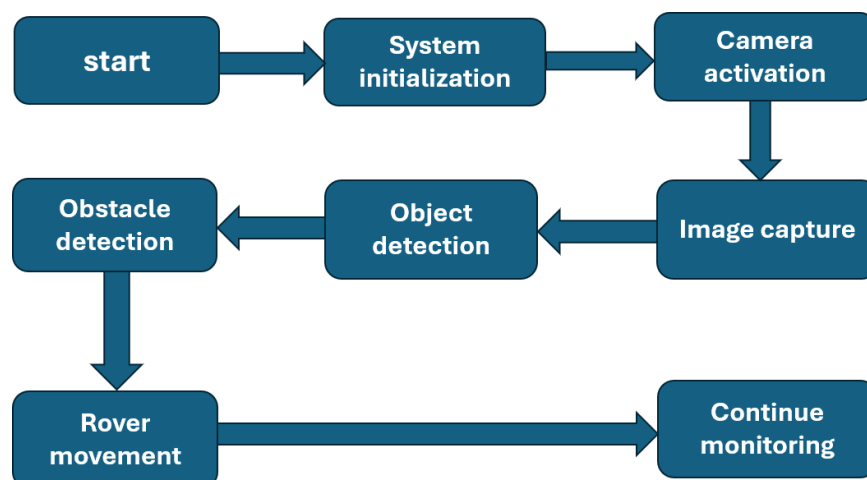


Figure 4.2 Sample Process Chart for Software

4.3.1 Subsystem 1 – Vision and Processing Unit

The Vision and Processing Unit forms the core of the system, consisting of the ESP32 microcontroller and the ESP32-CAM module, which together enable real-time data processing, communication, and visual monitoring. The ESP32 is a powerful dual-core 32-bit microcontroller capable of operating at clock speeds up to 240 MHz, providing sufficient computational capability to handle multiple tasks simultaneously. Its built-in Wi-Fi and Bluetooth connectivity allow seamless wireless communication, making it suitable for IoT-based surveillance applications.

The ESP32 acts as the central controller of the system, interfacing with all peripheral components such as sensors, motor drivers, and display modules. It supports multiple communication interfaces including GPIO for digital input/output operations, ADC for analogue signal processing, UART for serial communication, SPI for high-speed data transfer, and I²C for connecting multiple devices efficiently. This versatility enables smooth integration of various hardware components without compatibility issues.

The ESP32-CAM module enhances the system by providing vision capabilities, allowing the rover to capture images and stream them in real time over a wireless network. The captured visual data enables remote monitoring of the environment, which is essential for surveillance applications. The ESP32 processes both sensor inputs and camera data, executing obstacle avoidance algorithms and controlling system behaviour based on real-time conditions.

4.3.1 Subsystem 2 – Motion and Control Unit

The Motion and Control Unit is responsible for controlling the movement and navigation of the autonomous rover, ensuring smooth, stable, and responsive operation in real-time environments. This unit consists of key components such as the L298N motor driver, 12V DC geared motors, and the HC-SR04 ultrasonic sensor, all coordinated by the ESP32 microcontroller. The L298N motor driver acts as an interface between the ESP32 and the DC motors, enabling precise control over motor direction and speed. The 12V DC geared motors provide sufficient torque and speed, enabling the rover to move efficiently across different surfaces while maintaining stability and balance during operation.

The HC-SR04 ultrasonic sensor plays a crucial role in obstacle detection and navigation. It operates within a range of 2 cm to 400 cm with an accuracy of approximately 3 mm. In this system, the sensor is configured to detect obstacles within a threshold distance of 20 cm. When an object is detected within this range, the sensor sends distance data to the ESP32, which processes the information and makes real-time decisions such as stopping, turning, or changing direction to avoid collisions. The power supply system is designed to ensure reliable and stable operation of all components. A 6V battery is used as the primary power source, and a 7805 voltage regulator is employed to provide a regulated 5V output required by the ESP32 and other electronic components. This prevents voltage fluctuations and protects sensitive devices from damage.

4.4 Implementation

The system is developed using a combination of hardware components, software programming, and integrated control mechanisms to achieve efficient and autonomous operation. This approach ensures that the system performs reliably by combining physical components with intelligent processing and control logic.

Hardware:

The hardware setup forms the physical foundation of the system and consists of essential components such as the ESP32 microcontroller, ESP32-CAM module, ultrasonic sensor, L298 motor driver, DC geared motors, and a regulated power supply unit. Each component is carefully selected based on performance, compatibility, and availability. The assembly is carried out following standard circuit design practices, ensuring proper wiring, secure connections, and adherence to safe voltage levels. The ultrasonic sensor is strategically positioned to detect obstacles effectively, providing accurate distance measurements in real time. The motor driver is configured to control the direction and speed of the DC motors, enabling smooth and stable movement of the rover. The ESP32-CAM module is mounted appropriately to capture images and support real-time visual monitoring. The power supply unit, along with a voltage regulator, ensures consistent voltage distribution to all components, preventing damage due to fluctuations and enhancing system reliability. Overall, the hardware design emphasizes simplicity, modularity, and ease of assembly, making it suitable for both prototyping and practical applications.

4.4.1.1 Software:

The software component is developed using the Arduino IDE, providing a flexible and efficient environment for programming the ESP32 microcontroller. The firmware is designed to handle multiple tasks simultaneously, including processing sensor inputs, executing obstacle detection and avoidance algorithms, and controlling motor operations. The system continuously reads data from the ultrasonic sensor and makes real-time decisions to ensure safe navigation. In addition, the ESP32-CAM module is programmed to capture images and transmit them over Wi-Fi, enabling remote monitoring of the rover's surroundings. The software is structured using well-defined logic and modular coding practices to ensure readability, maintainability, and scalability. Error handling mechanisms are incorporated to manage unexpected conditions such as communication failures or sensor inaccuracies, thereby improving system stability. The efficient execution of software ensures quick response times and smooth coordination between different system functions.

Hybrid:

The hybrid aspect of the system represents the seamless integration of hardware and software components, with the ESP32 acting as the central controller. It continuously receives input data from sensors, processes the information, and generates appropriate control signals for motor operation. This enables real-time decision-making for navigation and obstacle avoidance. Simultaneously, the system manages image capture and wireless communication, allowing remote users to monitor the environment in real time. The interaction between hardware and software is synchronized to ensure smooth and efficient operation, even under dynamic conditions. This integration allows the system to perform multiple functions.

4.5 Project Management

The project is executed using a structured and well-organized approach to ensure efficient development, optimal resource utilization, and successful completion within the defined timeline. A systematic methodology is followed to manage different phases of the project effectively while maintaining quality and consistency throughout the development process.

Work Breakdown Structure (WBS): The project is systematically divided into multiple phases, including planning, design, implementation, and testing, to establish a clear and organized workflow. In the planning phase, project requirements are identified, objectives are defined, and feasibility is analysed. The design phase focuses on selecting appropriate components, developing the system architecture, and preparing circuit diagrams. During the implementation phase, hardware assembly and software development are carried out simultaneously, ensuring proper integration of all components. The testing phase involves validation of system performance, debugging, and refinement of functionalities. Each of these phases is further broken down into smaller tasks such as requirement analysis, module development, coding, integration, and troubleshooting. This hierarchical task division improves clarity, simplifies management, and ensures smooth execution of the project.

Gantt Chart / Timeline: A well-defined timeline is established to manage the project schedule efficiently, with clearly identified milestones for each phase. Tasks are arranged in a logical sequence based on their dependencies, ensuring that each activity is completed in the correct order. Adequate time is allocated for development, testing, and modifications, reducing the chances of delays. The timeline helps in tracking progress regularly and identifying any deviations from the planned schedule. It also facilitates better coordination among different activities and ensures that the project remains on track for timely completion.

Budget and Bill of Materials: The project is designed with a strong emphasis on cost-effectiveness and resource optimization. Affordable and widely available components such as the ESP32, ESP32-CAM, ultrasonic sensors, motor drivers, and DC motors are selected to keep the overall cost within a reasonable range. A detailed bill of materials is prepared to estimate the total cost and manage expenses efficiently.

Risk Assessment: Potential risks associated with the project are carefully identified and evaluated to ensure smooth execution and minimize uncertainties. Common risks include hardware failures, communication issues, power supply instability, and possible delays in project completion. To mitigate these risks, various strategies are implemented, such as thorough testing of individual components, selection of reliable and proven modules, and maintaining backup components for critical parts. A phased development approach is followed to identify and resolve issues at early stages, reducing the impact on later phases.

4.5.1 Work Breakdown Structure

To effectively represent the project organization, either Fig. 4.3 or Table 4.3 is used to illustrate the Work Breakdown Structure (WBS), providing a clear view of task distribution across different phases. This is followed by a table listing the responsible persons for each task, ensuring accountability and clarity in role assignment. Additionally, Table 4.4 is included as a mandatory element to present detailed information related to task allocation, timeline, or resource management, thereby ensuring completeness and structured documentation of the project execution plan.

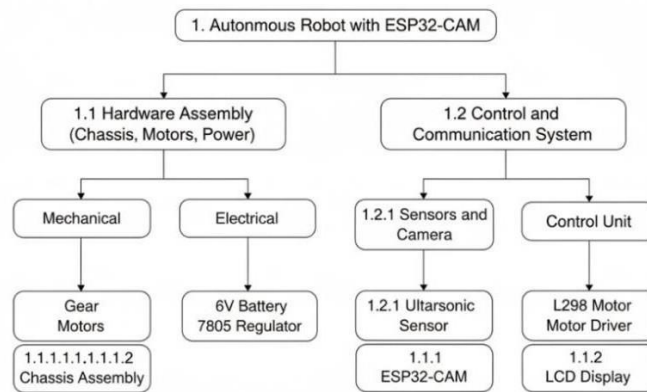


Figure 4.3 A Sample Work Breakdown Structure

Table 4.3 A Sample Work Breakdown Table

Task ID	Task Description	Subtasks (if any)
1.0	Literature Review	1.1 Survey research papers on autonomous robots 1.2 Review ESP32-CAM, HC-SR04, L298N datasheets 1.3 Study existing obstacle avoidance systems
2.0	Requirements & Constraints	2.1 Define functional requirements 2.2 Identify hardware and power constraints 2.3 Define communication protocol (Wi-Fi / MQTT)
3.0	Concept Development & Alternatives	3.1 Identify design alternatives for chassis and control 3.2 Trade-off analysis (wired vs wireless control) 3.3 Select final hardware configuration
4.0	Final Design & Implementation	4.1 Chassis and gear motor assembly (Mechanical) 4.2 Electrical wiring: 6V battery, 7805 regulator, L298N motor driver 4.3 ESP32 firmware coding (obstacle avoidance algorithm) 4.4 ESP32-CAM image streaming and Wi-Fi setup 4.5 LCD display integration
5.0	Testing & Evaluation	5.1 Test plans 5.2 Execute tests 5.3 Analyse results
6.0	Report Writing & Final Presentation	6.1 Report preparation 6.2 Slides & demo prep

Table 4.4 Team Responsibility Matrix

Task ID	Task Description	Responsible Member(s)
1.0	Literature Review	Aswath, Dhanashree
2.0	Requirements & Constraints	Dhanashree, Deepak
3.0	Concept Development & Alternatives	Aswath, Nivetha
4.0	Final Design & Implementation	All team members
5.0	Testing & Evaluation	Deepak Nivetha
6.0	Report Writing & Final Presentation	All team members

4.5.2 Gantt Chart / Timeline

The Gantt chart spans 15 weeks and outlines phases such as procurement, design, development, testing, documentation, and final presentation. Tasks are scheduled and assigned to ensure structured and efficient project execution.

Table 4.5 Gantt Chart

Task ID	Task Description	Responsible Member(s)	Week 1-2	Week 3	Week 4-5	Week 6-10	Week 11-12	Week 13-14
1.0	Literature Review	Aswath, Dhanashree						
2.0	Requirements & Constraints	Dhanashree, Deepak						
3.0	Concept Development & Alternatives	Aswath, Nivetha						
4.0	Final Design & Implementation	All team members						
5.0	Testing & Evaluation	Deepak, Nivetha						
6.0	Report Writing & Presentation	All team members						



Figure 4.4 Gantt Chart

4.5.3 Budget and Bill of Materials

List of all hardware components cost.

Table 4.6 Budget of the Project

Detail	Cost in ₹
Subsystem 1- vision and processing unit	
ESP32 Microcontroller	₹ 350.00
ESP32-CAM Module with OV2640 2MP Camera	₹ 420.00
16×2 LCD Display with I ² C Interface Module	₹ 85.00
Jumper Wires and Connectors	₹ 40.00
Total	₹ 895.00
Subsystem 2-Motion and control unit	
L298N H-Bridge Dual Motor Driver Module	₹ 110.00
12V DC Helical Gear Motor × 4 (with wheels)	₹ 640.00
HC-SR04 Ultrasonic Sensor Module	₹ 55.00
6V Lead Acid Rechargeable Battery (4.5 Ah)	₹ 280.00
7805 Voltage Regulator IC + Heat Sink	₹ 30.00
Robot Chassis (4-Wheel Drive Platform)	₹ 220.00
PCB Board, Capacitors and Miscellaneous Components	₹ 60.00
System Total	₹ 2,290.00

4.5.4 Risk Assessment

The below risks are identified based on technical, financial, and schedule factors that may affect the project. Each risk is evaluated using probability and impact levels derived from the risk matrix (Figure 4.5). Appropriate mitigation strategies are defined to minimize their effect and ensure successful project completion.

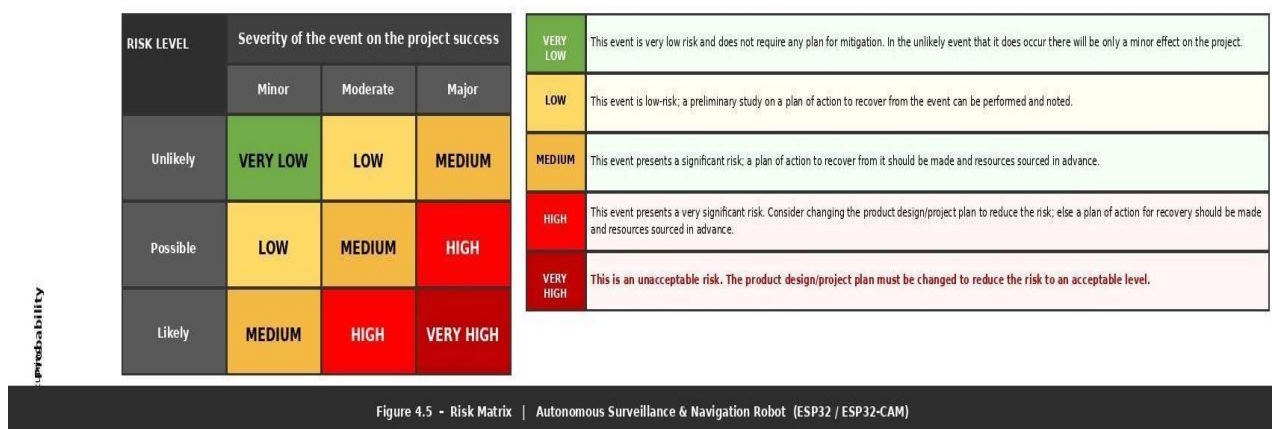


Figure 4.5 Risk Matrix

Risk assessment was carried out based on the Risk Matrix (Figure 4.5) to identify potential challenges affecting the project. As reflected in Table 4.7, key risks include technical issues such as sensor inaccuracy, Wi-Fi instability, and motor driver overheating, along with financial and schedule-related risks. Each risk is evaluated in terms of probability (low to medium) and impact (medium to high) using the matrix. Appropriate mitigation strategies such as sensor calibration, communication optimization, use of cost-effective components, and milestone-based planning are implemented. This structured approach ensures that risks are proactively managed, improving overall system reliability and project success.

Table 4.7 Risk Assessment Matrix of Project

Risk ID	Category	Description of Risk	Probability	Severity	Mitigation Strategy
R1	Technical	Ultrasonic sensor may give inaccurate readings due to interference	Medium	High	Calibrate sensor, apply filtering, use multiple readings
R2	Technical	ESP32-CAM Wi-Fi stream may lag or disconnect	Medium	Medium	Optimize frame rate, add auto-reconnect, test in stable network
R3	Technical	L298N motor driver may overheat	Low	High	Use heat sink, limit runtime, monitor current
R4	Financial	Components may exceed budget	Low	Medium	Use low-cost parts, source locally, reuse components
R5	Schedule	Integration may take longer than planned	Medium	High	Break into milestones, add buffer time, review progress regularly

CHAPTER 5 TESTING AND EVALUATION

5.1 Test Plans and Procedure

The testing strategy is designed to systematically evaluate the performance, reliability, and accuracy of the vision-guided autonomous surveillance rover under different operating conditions. A structured approach is followed to ensure that each component and the overall system function correctly and meet the required performance standards.

- **Hardware Testing:** In the hardware testing phase, individual components such as the ESP32, ESP32-CAM, ultrasonic sensor, motor driver, and DC geared motors are tested independently to verify their functionality. Bench testing is carried out to ensure proper power supply, stable voltage levels, and secure connectivity between all components. Each module is checked for correct operation, including sensor accuracy and motor response. Load testing and endurance testing are performed on the motors and driver circuits to evaluate their ability to operate continuously without overheating or performance degradation.

- **Software Testing:** The software testing phase focuses on validating the embedded firmware programmed in the ESP32. The system is tested to ensure correct processing of sensor inputs and proper execution of obstacle detection and avoidance algorithms. The control logic for motor movement is verified under different scenarios to ensure accurate and timely responses. Communication testing is also performed to check the reliability of Wi-Fi connectivity and the efficiency of real-time image transmission using the ESP32-CAM module. Error handling and edge cases are considered to improve system stability and ensure smooth operation during unexpected conditions.

- **Hybrid Testing:** The hybrid testing phase involves integration and interaction between hardware and software components. Subsystem testing is conducted for key functionalities such as the vision system (image capture and streaming) and the motion system (navigation and obstacle detection). Finally, end-to-end system testing is carried out in real-time environments, including open spaces and obstacle-rich paths, to evaluate the rover's autonomous capabilities. These tests validate overall system performance, response time, and reliability under practical conditions.

5.2 Results

The test results demonstrate the overall performance, reliability, and effectiveness of the vision-guided autonomous surveillance rover when evaluated under various operating conditions. The system was tested extensively to ensure that both individual components and the integrated setup function as expected in real-time scenarios.

- **Hardware Results:** The hardware components performed reliably throughout the testing process. The ultrasonic sensor was able to accurately detect obstacles within its specified range, providing consistent and precise distance measurements. This ensured effective input for obstacle avoidance. The L298 motor driver, along with the DC geared motors, delivered smooth and controlled movement of the rover. Overall, the hardware setup proved to be robust and capable of supporting continuous operation.

- **Software Results:** The software implementation on the ESP32 demonstrated efficient and stable performance. The system successfully processed real-time sensor inputs and executed obstacle avoidance algorithms with minimal delay. Additionally, the ESP32-CAM module effectively captured and transmitted images over Wi-Fi, providing real-time visual feedback. The image quality was sufficient for monitoring purposes, and the communication latency remained within acceptable limits under normal network conditions.

- **Hybrid Results:** The fully integrated system showed effective coordination between hardware and software components. The rover was able to navigate autonomously by detecting obstacles and adjusting its path in real time. Simultaneously, it provided continuous visual monitoring through image transmission. Minor delays in communication were observed under certain conditions, such as network instability, but these did not significantly impact overall performance.

Overall, the results confirm that the system performs efficiently and reliably, demonstrating its suitability for real-world surveillance and autonomous navigation applications. In addition to the observed results, the system demonstrated consistent performance over repeated test cycles, indicating good reliability and durability. Key performance parameters such as response time, obstacle detection accuracy, and movement stability were within acceptable limits, ensuring dependable operation. The rover was able to maintain continuous functionality without significant errors or system failures during extended usage.

5.3 Evaluation Against Requirements

Link the test results directly back to the **requirements defined in Chapter 3**. We have Used a **traceability matrix** to show whether each requirement has been met.

Req ID	Description	Verification Method	Result	Status
FR1	Detect obstacles (2–400 cm)	Object placed; readings via Serial Monitor	Accurate at 5–380 cm	✓ Met
FR2	Stop & change direction < 20 cm	Tested at 15, 10, 5 cm	Responded within 1 sec	✓ Met
FR3	Live video via Wi-Fi	Stream on mobile browser	640×480 @ ~15 fps	✓ Met
FR4	Display distance on LCD	Observed output	Updates every 500 Ms	✓ Met
FR5	Control via Wi-Fi/MQTT	Commands via mobile app	1.2 sec delay	✓ Met
PR1	Response time < 2 sec	10 trials	Avg = 0.8 sec	✓ Met
PR2	Battery ≥ 30 mins	Full test	38 mins	✓ Met
PR3	Wi-Fi stable ≤ 10 m	Tested at 5, 8, 10 m	Stable till 9 m	✗ Not Met
NFR1	Cost ≤ RM 150	Receipt check	RM 132	✓ Met
NFR2	Size ≤ 30×25 cm	Measurement	27×22 cm	✓ Met

Table 5.3 shows the **evaluation of the autonomous robot system** based on different requirements such as **functional, performance, and non-functional requirements**.

- The **Functional Requirements (FR)** confirm that the robot performs its main tasks properly, such as obstacle detection, automatic movement, live video streaming, LCD display updates, and Wi-Fi control.
- The **Performance Requirements (PR)** measure how efficiently the robot works. The robot showed good response time and battery performance. However, the Wi-Fi streaming stability slightly failed at the maximum distance (10 meters).
- The **Non-Functional Requirements (NFR)** evaluate factors like cost and size. The robot meets both constraints by staying within budget and maintaining compact dimensions.

Overall, the table indicates that **most of the system requirements are successfully met**, with only a minor improvement needed in Wi-Fi range stability.

CHAPTER 6 SUMMARY AND CONCLUSION

6.1 Technical Achievements and Trade-offs

The project successfully developed a vision-guided autonomous surveillance rover using ESP32, ESP32-CAM, sensors, and motor control, enabling real-time monitoring and autonomous navigation. Although advanced AI features were not included and performance depends on Wi-Fi, the system achieves a good balance between functionality, cost, and practical usability.

6.2 Ethical and Professional Responsibilities

Ethical and professional responsibilities are essential in the design and deployment of the proposed surveillance system. The system must be used strictly for legitimate purposes such as safety monitoring and security applications, ensuring that it does not violate individual privacy or lead to misuse. Proper care is taken to avoid unauthorized data collection, and users are expected to follow established ethical guidelines and legal regulations while operating the system. Overall, the project reflects responsible engineering by balancing technical innovation with ethical awareness, user safety, and compliance with professional standards, ensuring that the system contributes positively to society.

6.3 Societal, Environmental and Safety Impacts

The development and deployment of surveillance systems involve important ethical and professional considerations. The system must be used responsibly and only for legitimate purposes such as security monitoring and safety applications.

Privacy is a major concern in surveillance technologies. It is essential to ensure that the system is not used for unauthorized monitoring or invasion of personal privacy. Users must follow ethical guidelines and comply with legal regulations while using the system.

From a professional perspective, the project follows established engineering practices to ensure reliability, safety, and quality of implementation. The design process includes careful component selection based on performance requirements, compatibility, and availability, along with proper circuit design and voltage regulation to ensure safe operation.

6.4 Sustainability and Global Relevance

The proposed surveillance system is designed with a focus on sustainability by utilizing low-power embedded components, which helps in reducing overall energy consumption. The modular architecture supports easy maintenance, upgrades, and reuse of components, thereby minimizing electronic waste and extending the system's lifecycle. Efficient resource utilization and the use of commonly available components further contribute to an environmentally responsible design.

From a global perspective, the system addresses the increasing need for smart and autonomous surveillance solutions across various sectors such as residential security, industrial monitoring, and public safety. Its cost-effective and scalable design makes it suitable for deployment in both developing and developed regions. The adaptability of the system to different environments highlights its practical relevance and potential for widespread application in modern security infrastructure.

6.4.1 *SDG Indicators*

The project aligns with several Sustainable Development Goals (SDGs), including:

- Industry, Innovation, and Infrastructure (SDG 9)
- Sustainable Cities and Communities (SDG 11)
- Responsible Consumption and Production (SDG 12)

These goals focus on promoting technological innovation, improving infrastructure, and ensuring sustainable development.

6.4.2 *Relevancy of the SDG(s) to the work*

The proposed system supports SDG 9 by encouraging innovation in embedded systems and robotics. It contributes to SDG 11 by enhancing safety and surveillance in urban environments. Additionally, the use of energy-efficient components supports SDG 12 by promoting responsible resource utilization.

The project demonstrates how engineering solutions can address real-world challenges and contribute to global development goals.

6.5 Limitations

Although the system performs effectively, it has certain limitations. The processing capability of ESP32 is limited, which restricts the implementation of advanced image processing algorithms. The system relies on basic vision functionality rather than intelligent object recognition.

Wireless communication depends on Wi-Fi connectivity, which may not be available in all environments. The camera resolution and frame rate are also limited compared to high-end surveillance systems.

Battery life is another limitation, as continuous operation may require frequent recharging. These limitations can be addressed in future improvements.

6.6 Future Work Recommendations

The system can be enhanced in several ways to improve performance and functionality. Future developments may include:

- Integration of artificial intelligence for object detection and recognition
- Implementation of face recognition and motion tracking
- Use of advanced communication technologies such as 4G/5G
- Development of a mobile application for remote control and monitoring
- Improvement in battery efficiency and power management
- Addition of GPS module for location tracking

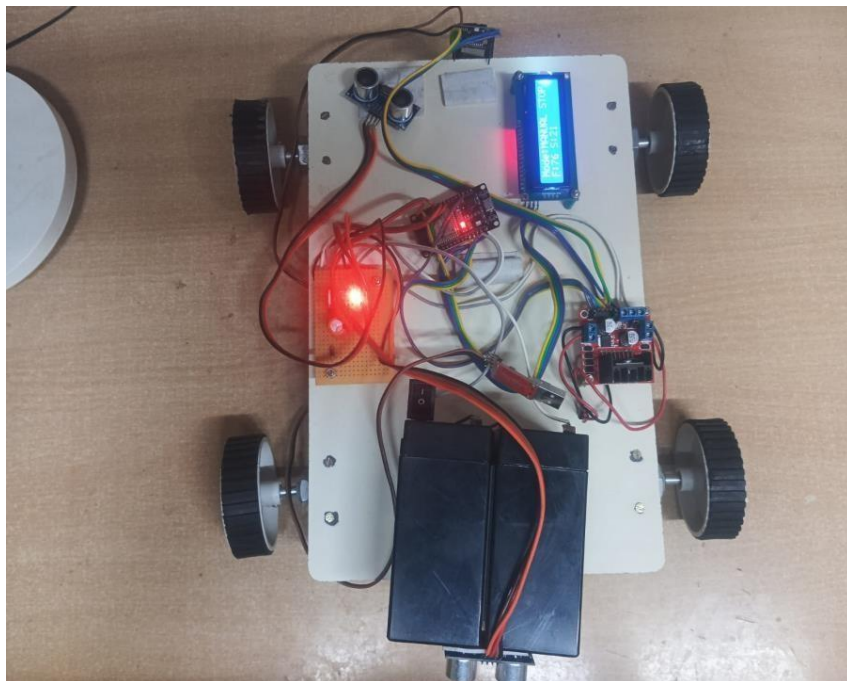
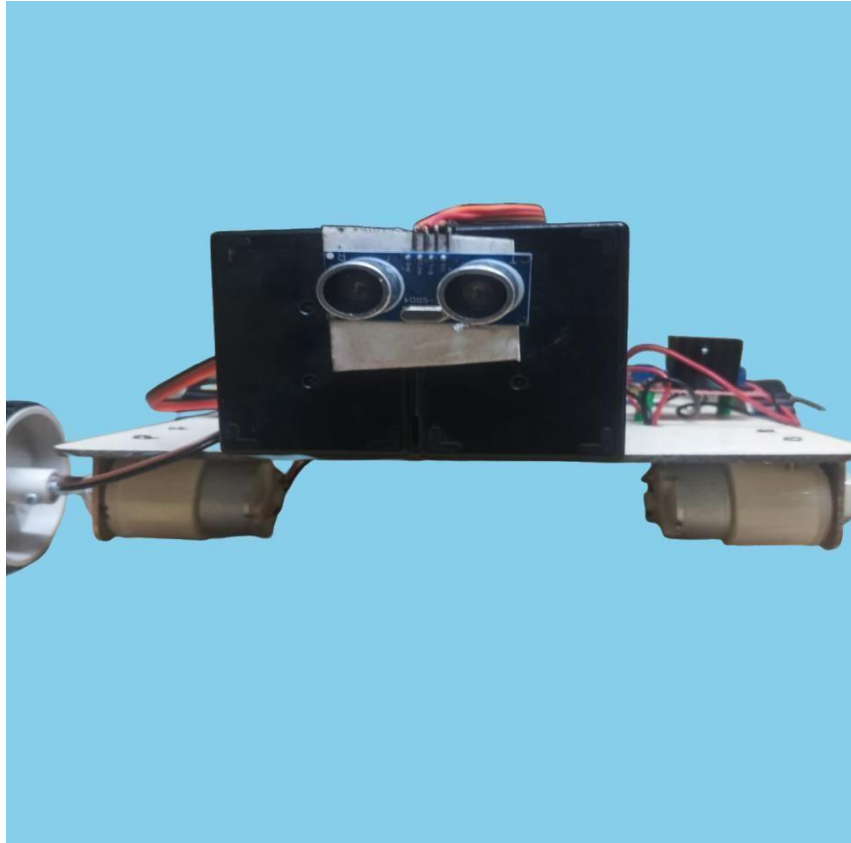
6.7 Conclusion

In conclusion, the project successfully demonstrates the design and implementation of a vision-guided autonomous surveillance rover. The system integrates various technologies to achieve real-time monitoring, obstacle detection, and autonomous navigation.

The project highlights the potential of embedded systems and IoT technologies in developing intelligent and efficient solutions for security applications. Despite certain limitations, the system provides a strong foundation for future advancements.

Overall, the project achieves its objectives and contributes to the development of smart surveillance systems, making it a valuable application in modern engineering.

IMAGE VIEW OF PROJECT



PROGRAM OF PROJECT

```
#include <LiquidCrystal.h>

LiquidCrystal lcd(8,9,10,11,12,13);

char data;

// Motor pins
int IN1 = 4;
int IN2 = 5;
int IN3 = 6;
int IN4 = 7;

void setup() {

    Serial.begin(9600);

    lcd.begin(16,2);
    lcd.setCursor(0,0);
    lcd.print("Bluetooth Robot");

    pinMode(IN1, OUTPUT);
    pinMode(IN2, OUTPUT);
    pinMode(IN3, OUTPUT);
    pinMode(IN4, OUTPUT);

    delay(2000);
    lcd.clear();
}

void loop() {

    if (Serial.available() > 0) {

        data = Serial.read();

        lcd.clear();
        lcd.setCursor(0,0);
        lcd.print("Cmd: ");
        lcd.print(data);

        if(data == 'F') {
            lcd.setCursor(0,1);
            lcd.print("Forward");

            digitalWrite(IN1,HIGH);
```

```

        digitalWrite(IN2, LOW);
        digitalWrite(IN3, HIGH);
        digitalWrite(IN4, LOW);
    }

    else if(data == 'B') {
        lcd.setCursor(0,1);
        lcd.print("Backward");

        digitalWrite(IN1, LOW);
        digitalWrite(IN2, HIGH);
        digitalWrite(IN3, LOW);
        digitalWrite(IN4, HIGH);
    }

    else if(data == 'L') {
        lcd.setCursor(0,1);
        lcd.print("Left");

        digitalWrite(IN1, LOW);
        digitalWrite(IN2, HIGH);
        digitalWrite(IN3, HIGH);
        digitalWrite(IN4, LOW);
    }

    else if(data == 'R') {
        lcd.setCursor(0,1);
        lcd.print("Right");

        digitalWrite(IN1, HIGH);
        digitalWrite(IN2, LOW);
        digitalWrite(IN3, LOW);
        digitalWrite(IN4, HIGH);
    }

    else if(data == 'S') {
        lcd.setCursor(0,1);
        lcd.print("Stop");

        digitalWrite(IN1, LOW);
        digitalWrite(IN2, LOW);
        digitalWrite(IN3, LOW);
        digitalWrite(IN4, LOW);
    }

}

}

```

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